



- 2025 -01.webp
- 2025 -02.webp
- 2025 -03.webp
- 2025 -04.webp
- 2025 -05.webp

$$\begin{array}{ccccccc} & a & \triangle ABC & BF & AC & BF=b & D \quad BF \\ AD & AD & \triangle ADE & EF & \triangle AEF & & \\ \frac{1}{2}a+\frac{2}{3}b & \frac{1}{2}a+b & a+\frac{1}{2}b & \frac{3}{2}a & & & \end{array}$$

$$x,y \qquad |x|+4|y|=1 \qquad m=y-4x$$

$$\cdots + G(2019) = 90 \qquad \begin{array}{cc} a & G(x) = a - x + |x - a| \\ & a \end{array} \qquad G(1) + G(2) + G(3) +$$

$$\begin{array}{ccccccc} & AB & A & (2,1) & B & (4,1) \\ C & AB & B & C & 90^\circ & D & y = \\ \frac{k}{x} (x > 0) & D & k & & & & \end{array}$$

$$\begin{array}{ccccccc} (x-a)(x-b)-x=0 & c,d & (x-c)(x-d)+x=0 \\ a,b & -a,-b & c,d & -c,-d & & & \end{array}$$

$$\cos \angle BOC = \frac{3}{5} \qquad \begin{array}{c} ABOC \\ C \end{array} \qquad \begin{array}{c} O \\ (a,4) \end{array} \qquad \begin{array}{c} BO \\ y = \frac{k}{x} \end{array} \qquad \begin{array}{c} x \\ -\frac{50}{3} \end{array}$$

$$\begin{array}{c} AO \quad D \quad BD \quad BD \perp x \quad k \\ -\frac{25}{2} \quad -12 \quad -\frac{25}{4} \end{array}$$

$$\begin{array}{c} x \quad [x] \quad x \quad [2] = 2, [1.2] = 1 \\ \left[\frac{4x+a}{3} \right] = 2 \quad a \quad 0 < a < 1 \quad 2 \leq \\ a < 3 \quad 0 < a \leq 1 \quad 2 \leq a < 3 \quad 0 < a < 1 \quad 2 \leq a < 5 \quad 0 < a \leq 1 \\ 2 < a \leq 5 \end{array}$$

$$a,b \qquad \frac{a}{2^4+4^3} + \frac{b}{2^4+5^3} = 1 \qquad \frac{a}{3^4+4^3} + \frac{b}{3^4+5^3} = 1 \qquad a+b =$$

$$\begin{array}{c} ABCD \quad AB = 1 \quad BC = 2 \quad AE \quad \angle BAD \quad F \quad AE \\ M \quad DF \quad BM \quad BM \end{array}$$

$$\begin{array}{c} 2x(|x|-2) = 1 \\ y = 2x(|x|-2) \qquad y = 1 \\ 2x(|x|-2) = 1 \\ x \leq 0 \quad y = -2x^2 - 4x \quad x > 0 \quad y = \rule{1cm}{0.4pt} \\ x \leq 0 \qquad \qquad \qquad x > 0 \\ y = 1 \qquad 2x(|x|-2) = 1 \\ x \quad 2x(|x|-2) = m \\ m \quad \rule{1cm}{0.4pt} \end{array}$$

$$\sqrt{a+2\sqrt{a-1}} + \sqrt{a-2\sqrt{a-1}}$$

$$\begin{array}{c} \triangle ABC \quad \angle BAC = 60^\circ \quad AB = 2AC \quad P \quad \triangle ABC \quad PA = \\ \sqrt{3} \quad PB = 5 \quad PC = 2 \quad \triangle ABC \end{array}$$

$$\begin{array}{ccccccc}
 & \text{Rt}\triangle ABC & \angle ACB = 90^\circ & AB = 5 & \sin A = \frac{4}{5} & P & \\
 BC & PE \perp AB & E & P & PC & PE & \\
 & CQ & AB & D & AD & CP = x & \\
 \triangle PCQ & y & y & x & & C & \\
 CF \perp AB & F & PF & QF & \triangle PQF & PF & \\
 CP & & & & & &
 \end{array}$$

$$\begin{array}{ccccccc}
 O & AC & P & BC & AP & Q & \textcircled{1} \quad \tan \angle BAP = \frac{1}{3} \\
 BP & \textcircled{2} & \frac{CQ}{AP} & & & &
 \end{array}$$

$$\begin{array}{l}
 CQ \cdot AP = AC \cdot CP \quad CP = n \quad AP = \sqrt{64 + n^2} \\
 64 + n^2 \geq 16n
 \end{array}$$